

2 **Supplementary Methods**

3 **S1 Formal validation route**

4 All validation analyses followed the same three-tier architecture used by the software. A
5 query profile was first represented in logCPM- or logTPM-compatible expression space
6 and then projected into Bo2023-like VSD space for Network scoring only. The top three
7 Networks formed the candidate beam. Resolution-group and exact-region candidates
8 were then reranked within that beam using local logCPM-compatible expression. Fold-
9 local Network model construction differed between the LOSO and LOMO validation
10 designs as documented in S7, but projected VSD remained restricted to Network beam
11 generation in both. This design separates broad candidate generation from fine regional
12 interpretation.

13 **S1a Reference choice and cross-species interpretation**

14 The choice of a macaque reference is methodological rather than a claim that macaque
15 exact regions are directly interchangeable with human exact regions. Bo2023 provides a
16 controlled primate RNA-seq reference with 819 samples, repeated donors and a dense set
17 of cortical and subcortical labels that can be grouped into a stable Network, resolution-
18 group and exact-region hierarchy. That hierarchy is essential for the tool: Network Top3
19 forms the primary beam, resolution groups provide the main region-level endpoint when
20 local expression is not uniquely resolvable and exact-region output is deliberately
21 exploratory. Human datasets are then used as transfer tests. AHBA supports mapped-
22 label validation because its normal human brain labels can be harmonized to a subset of

23 the macaque-derived hierarchy. Tumour and biofluid datasets support only coarser
24 consistency, coverage or domain-transfer stress tests. This design avoids overfitting a
25 sparse human validation label space while still testing whether the primate-derived signal
26 transfers to real human expression matrices.

27 Important primate-to-human neuroanatomical differences remain. Human prefrontal
28 association cortex occupies a different proportional and functional landscape from
29 macaque prefrontal cortex, and language-dominant perisylvian systems do not have a
30 one-to-one macaque homologue. These differences are why the model reports
31 harmonized candidate rankings and resolution limits rather than asserting direct macaque-
32 to-human exact-region equivalence.

33 **S2 Internal validation design**

34 Internal validation used the Bo2023 macaque brain reference. Two settings were
35 evaluated. Leave-one-sample-out validation tested whether the route could recover the
36 held-out sample label when other samples from the reference remained available. Leave-
37 one-monkey-out validation held out all samples from one animal at a time and therefore
38 tested donor-level generalization. Network metrics included all 819 samples because
39 every held-out sample retained a supported Network label. Resolution-group and exact-
40 region metrics required the truth region to remain represented in the corresponding
41 training fold. Five LOSO samples and seven LOMO samples did not meet that region-
42 reference requirement and were excluded only from region-level denominators. Top1,
43 Top3 and median true-rank were used to distinguish exact calls from candidate-list
44 recovery.

S2a Region-ranking score

The locked hybrid route is a constrained hierarchical ranker. The first level uses projected-VSD Network evidence only to define a small candidate beam. The second and third levels do not reuse projected VSD for exact localization; they instead rerank candidate regions with logCPM-compatible local expression. In fold f , the training samples define Network centroids $c_k^{(f)}$, region centroids $\mu_r^{(f)}$ and a fold-local set of candidate regions $R_f(B)$. Given query x_i , the Network beam is $B_i = \text{Top3}_k(\text{corr}(P_f(x_i), c_k^{(f)}))$. Pairwise rescue is restricted to high-confusion Network pairs already present in B_i . For a candidate pair (a,b) , pair-specific gene set $G_{ab}^{(f)}$ is selected in the training fold and the pairwise scores are $q_a = \text{corr}(x_{Gab}, c_{a,Gab}^{(f)})$ and $q_b = \text{corr}(x_{Gab}, c_{b,Gab}^{(f)})$. If $q_a > q_b$, Network a may be ranked ahead of b , and conversely for b ; no Network outside the original Top3 beam can be introduced. The exact-region score for r in $R_f(B_i)$ is a z-score fusion of correlations over two nested local gene panels, $G50^{(f)}$ and $G100^{(f)}$. The locked value $\lambda=0.25$ weights the broader Top100 panel at 0.75 and the more selective Top50 panel at 0.25, reducing sensitivity to a very small marker set while retaining high-specificity local signal. This value was fixed in the hybrid route before the formal P0 validation refresh and was not optimized on AHBA, TCGA/BraTS or biofluid external datasets. This formulation separates the primary Network projection from downstream local expression evidence and prevents exact-region claims outside the candidate beam.

S3 External validation design

External validation was limited by the label resolution available in each dataset. AHBA human brain RNA-seq was used for mapped-label validation because its anatomical

68 labels could be harmonized to Network, resolution-group and a subset of exact-region
69 labels. TCGA/BraTS glioma tissue RNA-seq with MRI-derived labels was used only for
70 coarse anatomical consistency because its truth labels are human imaging labels, not
71 Bo2023 macaque exact-region identifiers. GSE189919 was used to test whether an
72 external matrix could be projected into the model gene space; it was not used for
73 accuracy estimation because patient-level anatomical truth was unavailable.

74 **S4 Label harmonization and allowed conclusions**

75 Label harmonization was performed only to the level supported by each dataset. AHBA
76 anatomical labels were mapped to the macaque-derived Network, resolution-group and
77 exact-region hierarchy where a supported mapping existed, and results are interpreted as
78 mapped-label transfer rather than direct anatomical equivalence. TCGA/BraTS labels
79 were derived from human MRI/tumour context and therefore support only coarse tumour-
80 tissue anatomical consistency. Biofluid datasets without patient-level anatomical truth
81 were not used for localization accuracy and are reported only as projection-feasibility or
82 transfer stress tests.

83 For gene-symbol handling, macaque Ensembl fallback identifiers beginning with
84 ENSMFAG were treated as valid macaque reference identifiers rather than spurious
85 symbols. They were retained in internal macaque marker summaries but were not used as
86 human gene symbols in human cfRNA or cross-species validation. Human-facing overlap
87 analyses used the humanized signature file
88 reports/bo2023_signature_genes_humanized_20260704.csv, in which 5324 of 8800
89 macaque signature rows (60.50%) mapped to a human gene symbol and 3476 remained
90 unmapped ENSMFAG identifiers. Human validation and coverage statements therefore

91 report the mapped humanized signature rather than counting unmapped macaque fallback
92 IDs as human genes.

93 The 60.50% row-level humanization rate is a transfer constraint, not a failure of the
94 macaque reference. It means that human-facing enrichment, coverage and cfRNA overlap
95 analyses are conducted on the mapped subset and may underrepresent macaque-specific
96 or poorly annotated genes. The formal AHBA validation is therefore interpreted at
97 harmonized label levels and with explicit gene-coverage reporting, while unmapped
98 ENSMFAG-prefixed rows are preserved only for macaque-reference transparency.

99 **S5 Model artifacts and projected-VSD projector**

100 The repository contains lightweight model artifacts in data/models/, including Network
101 model files, region-resolution dictionaries, route metadata and the reference-fitted
102 projector bo2023_reference_projector_linear_full.npz. The projector stores gene-wise
103 slope, intercept and clipping parameters fitted from reference data. At inference time, a
104 single query profile can be mapped into Bo2023-like projected-VSD space using these
105 fixed parameters without target-cohort labels or target-cohort distribution fitting.

106 Projected VSD is used only for Network Top3 beam generation; resolution-group and
107 exploratory exact-region reranking use logCPM-compatible local expression within that
108 beam.

109 The linear projection is an engineering compromise between computational efficiency
110 and expression-space alignment. It is appropriate for a fixed, single-sample compatible
111 transform, but it should not be interpreted as a complete model of cross-platform or
112 cross-species expression distortion. Strong nonlinear batch effects, platform-specific

gene-response curves or biological shifts not represented in the reference could reduce calibration; those cases should be handled through coverage, margin and entropy diagnostics and, where possible, dataset-specific validation rather than by assuming exact transfer.

S6 Example input/output

Synthetic public example files are provided in `submission_ready_assets/example_io/` as part of the v0.1.6 public submission release. They document accepted input columns, Network ranked candidates, three-tier JSON output, resolution-group ranked candidates, exact-region ranked candidates and the local generation script. The example files are format and reproducibility aids only; they are not biological validation samples.

S6a Software availability

The manuscript-associated software release is v0.1.6. The public GitHub repository is available at <https://github.com/wz7717/cfrna-brain-tracing>, and the archived release is cited in the manuscript using the Zenodo version DOI <https://doi.org/10.5281/zenodo.20780280>; the project concept DOI is <https://doi.org/10.5281/zenodo.20773674>.

S7 Model development and locked evaluation timeline

The three-tier architecture was fixed before generating the final submission validation tables. External AHBA, TCGA/BraTS and biofluid analyses were not used to select the final route; they were used only for mapped-label transfer evaluation, coarse tumour-tissue consistency assessment and biofluid transfer stress testing. The LOSO implementation used locked Network genes and correlation-ranked projected-VSD

Network Top3 candidates. The formal LOMO implementation rebuilt discriminative Network genes and a fold-local pairwise Top1 rescue within each training fold; this rescue could reorder Top1 but did not change the original Network Top3 candidate set. Both implementations used projected VSD only for the Network beam and logCPM-compatible local evidence for downstream reranking. The independent Network-only LOSO and LOMO analyses are reported separately as route-selection evidence.

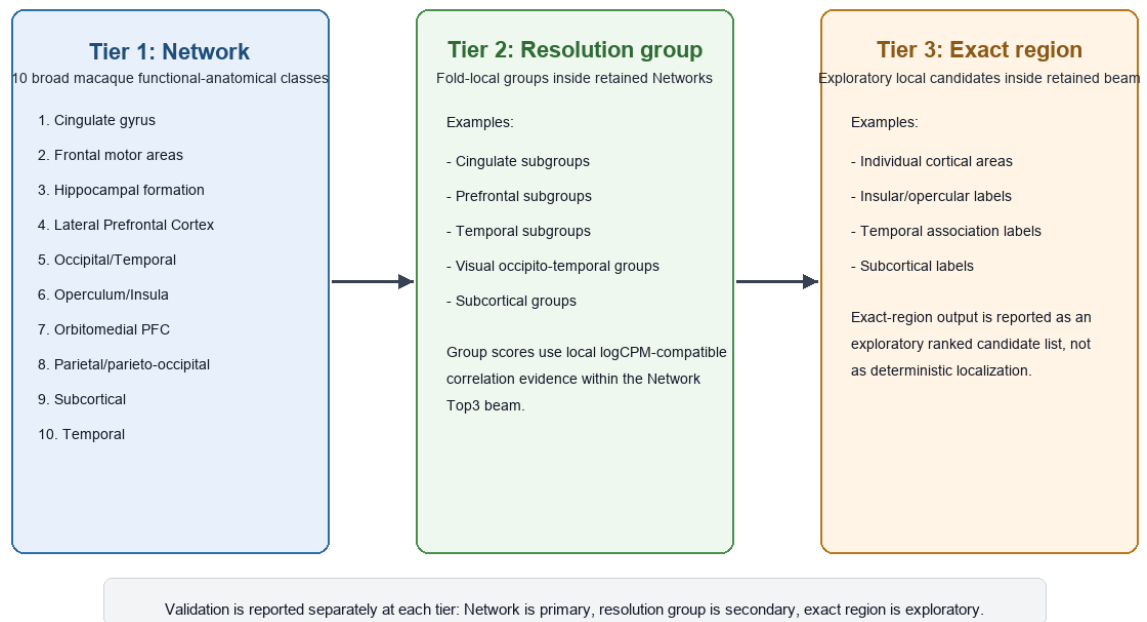
S8 Figures and tables

Figure source data and synthetic example input/output files are archived in the GitHub/Zenodo release rather than treated as separate journal supplementary files. Supplementary Tables are embedded below so that the formal supplementary PDF can be submitted as a single file containing Supplementary Methods, Supplementary Results and the supporting tables. Together, these materials document both how the route was tested and what result each validation setting supports. Supplementary Figure S1 visualizes the three-tier hierarchy from 10 Networks to resolution groups and exploratory exact regions; the source file is manuscript/figures_publication/Supplementary_Figure_S1_three_tier_hierarchy.svg. Resolution-group names in Supplementary Figure S1 are derived from the curated Bo2023 hierarchy and the same bo2023_region_resolution_groups.json artifact used by the formal route.

Supplementary Figure S1. Three-tier hierarchy used for Network, resolution-group, and exact-region validation reporting.

cfRNA-BrainTrace three-tier anatomical hierarchy

Projected-VSD Network Top3 beam followed by logCPM-compatible resolution-group and exact-region reranking



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157 S9 Evidence package provenance

158 The reviewer-response evidence package is organized into three tiers. P0 hard-evidence
159 files contain the formal validation metrics, Wilson confidence intervals, random
160 baselines, binomial tests, LOSO-vs-LOMO paired tests, confusion matrices, class-level
161 F1, denominator audits, marker-methodology audits, Network anatomy tables, resolution-
162 group hierarchy tables and AHBA mapping granularity summaries. P1 diagnostics
163 contain dual-space consistency, error-cascade and confidence-diagnostic analyses. P2
164 completeness files contain development/comparator materials, including same-field tool
165 positioning, simple ML baselines, test coverage, locked dependency files and random-
166 seed registry. P2 comparator files are retained as development, comparator, sensitivity or
167 diagnostic evidence; they are not merged into the formal three-tier hybrid validation
168 endpoint.

Supplementary Results

S1 Internal route selection

The first internal analysis evaluated Network-level candidate generation. Projected VSD achieved Network Top1/Top3 of 58.00%/91.58% in LOSO and 53.72%/91.33% in LOMO, exceeding logCPM baseline and native VSD at Network Top3. Direct exact-region scoring was lower and less stable, especially in LOMO. These results motivated the use of projected VSD for broad Network beam generation and logCPM-compatible expression for downstream local reranking.

S2 Formal internal three-tier validation

In the complete LOSO validation, the formal route achieved Network Top1/Top3 of 58.24%/92.19% across all 819 samples. Resolution-group and exact-region Top3 were 72.36% and 45.33% among 814 reference-supported samples. In complete LOMO validation, Network Top3 was 91.21% across all 819 samples, while resolution-group and exact-region Top3 were 69.09% and 42.36% among 812 reference-supported samples. The 92.19% LOSO Network Top3 value uses all 819 Network-evaluable samples as the denominator. Region-level LOSO metrics use 814 reference-supported samples because five samples lacked a truth-region reference after fold construction. Region-level LOMO metrics use 812 reference-supported samples because seven samples lacked a truth-region reference after fold construction. The earlier LOSO Network value of 92.38% was conditional on the 814 region-evaluable samples and is retained only as a legacy denominator inconsistency, not as the submission result. Median true-rank increased from Network to exact-region levels, consistent with decreasing

anatomical certainty at finer resolution. Resolution group is therefore the preferred region-level endpoint, while exact-region output is retained as a candidate ranking. Wilson confidence intervals (Wilson, 1927) and statistical tests were generated for the formal P0 evidence package. Network Top3 was 92.19% in LOSO (755/819; 95% CI 90.14-93.83%) and 91.21% in LOMO (747/819; 95% CI 89.07-92.96%). Resolution-group Top3 was 72.36% in LOSO (589/814; 95% CI 69.19-75.32%) and 69.09% in LOMO (561/812; 95% CI 65.83-72.17%). Exact-region Top3 was 45.33% in LOSO (369/814; 95% CI 41.94-48.77%) and 42.36% in LOMO (344/812; 95% CI 39.01-45.79%). Each endpoint exceeded its uniform random baseline by one-sided binomial testing, including exact-region LOSO Top3 versus 7.92% uniform random expectation ($p=4.43e-181$). Paired LOSO-vs-LOMO tests showed no significant Network Top3 decrease ($p=0.1686$), a significant resolution-group decrease ($p=0.0389$) and a borderline exact-region decrease ($p=0.0532$).

S3 External validation results

In AHBA, the formal three-tier hybrid route achieved Network Top1/Top3 of 74.68% (95% CI 68.73-79.83%)/94.42% (90.69-96.71%), resolution-group Top1/Top3 of 36.26% (27.13-46.51%)/67.03% (56.86-75.83%) and exact-region Top1/Top3 of 24.18% (16.54-33.90%)/42.86% (33.18-53.11%). Hybrid exact Top3 exceeded logCPM baseline (30.77%) and projected-VSD-only scoring (29.67%). These AHBA values are the main external validation result because the current manuscript uses AHBA as mapped-label human brain transfer, not as a historical route-development trace. In TCGA/BraTS, hybrid Network Top3 was 40.00% and broad-anatomy Top3 was 64.62%, supporting only coarse anatomical consistency. Against a nominal 30% Network Top3 reference

level, the TCGA/BraTS Network Top3 value of 26/65 has a one-sided normal-approximation p value of approximately 0.039; the more conservative exact binomial test gives $p=0.0548$. Against the endpoint-specific weighted random baseline of 20.28%, the exact one-sided binomial p value is $2.17e-4$. These values support only coarse tumour-tissue consistency and do not convert TCGA/BraTS into an exact localization validation set. GSE189919 overlapped 15,622/21,668 projector genes, corresponding to 72.10% gene-space coverage, supporting projection feasibility rather than source-localization accuracy.

S4 Dual-space and diagnostic analyses

Projected VSD and logCPM are not independent competing endpoints. Projected VSD defines the broad Network Top3 candidate beam, and logCPM then performs local reranking inside that beam. In the P1 diagnostic package, LOSO exact-region Top1 candidates were inside the retained Network beam in 100.00% of cases, LOSO exact-region Top3 candidates were all inside the beam in 100.00% of cases and no Top3 candidate occurred outside the retained beam. Conditional on a correct Network beam, LOSO exact-region Top3 was 49.07% and resolution-group Top3 was 78.32%; after a missed Network beam, LOSO exact-region recovery was 0.00%. These results support the intended cascade interpretation: Network misses are near-hard failures for downstream recovery, whereas remaining downstream misses after a correct beam reflect local anatomical ambiguity.

Diagnostic outputs were evaluated as confidence signals rather than as independent validation endpoints. Full Network scores provided top1_margin, top3_beam_margin, normalized score entropy and marker-overlap counts. For LOSO exact-region Top3, the

Top3 beam margin had AUC 0.613 for downstream hit/miss discrimination, whereas marker overlap was not testable in internal Bo2023 or AHBA validation because the relevant marker panels were fully covered. An AUC of 0.613 is only modestly above the random-discrimination value of 0.5, so score margin should not be used as a binary accept/reject decision rule; it is a weak confidence aid to be interpreted together with anatomical level, marker coverage, entropy and dataset scope. External biofluid matrices without anatomical truth therefore support only coverage and projection-feasibility checks. The GSE189919 projector overlap result is used in this limited sense: it indirectly tests whether a sparse external profile can enter the gene space, but it does not test cfRNA degradation robustness or localization accuracy.

Reported p values are per-endpoint and were not adjusted for multiple comparisons because the endpoints address distinct anatomical-resolution hypotheses: Network candidate recovery, resolution-group recovery, exploratory exact-region recovery and coarse external consistency.

S5 Comparator, sensitivity and reproducibility materials

Simple ML baselines were run as leave-one-monkey-out Network classifiers on Bo2023 VSD expression with training-fold feature selection. Top3 accuracies were 80.46% for 5-nearest-neighbour cosine classification, 83.27% for nearest centroid and 90.96% for random forest. These baselines are retained as comparator evidence only: they do not return resolution groups, exact-region candidate rankings, resolution-limit diagnostics or the formal three-tier route used by the software. Same-field comparison found no strict like-for-like public tool. The closest named comparators are CIBERSORTx and related cell-type deconvolution tools, TissueEnrich and tissue-expression enrichment tools, broad

cell-free transcriptome tissue-of-origin studies, and Allen/AHBA atlas-query workflows. cfRNA-BrainTrace should therefore be positioned as a resolution-aware hierarchical candidate-ranking and audit tool rather than a generic tissue deconvolution method.

The baseline comparison supports the formal route rather than replacing it. Random forest approaches the formal LOMO Network Top3 result at the broad Network level, but it does not provide the required hierarchical outputs, fold-local resolution-group reranking, exploratory exact-region candidate lists, score-margin/entropy diagnostics or explicit abstention boundaries. The formal hybrid route is therefore preferred for the manuscript because it is a validated end-to-end anatomical ranking workflow, whereas ML baselines are restricted comparator checks for broad Network classification.

Formal LOSO sensitivity analyses reran the same P0 three-tier pipeline with alternative local exact-region fusion weights. The locked route used $\lambda=0.25$, which gave the highest exact-region Top1 accuracy among the evaluated values; $\lambda=0.50$ gave a numerically higher Top3 value, while $\lambda=0.75$ was slightly lower than $\lambda=0.50$. Because the manuscript primary route had been locked before the formal validation refresh and because all three values gave similar median true-rank, $\lambda=0.25$ was retained as the prespecified formal route rather than reselected by post hoc Top3 optimization.

The engineering reproducibility package includes a pinned requirements-lock.txt, an environment.yml, a random-seed registry and coverage artifacts. These files address software reproducibility and reviewer traceability but do not alter the formal validation route.

S6 Cerebellum/posterior-fossa scope and cfRNA domain transfer

We intentionally avoid inferring cerebellar or posterior-fossa origin from the present model. The reason is not that these locations are biologically irrelevant, but that the current locked evidence chain lacks a validated cerebellum-inclusive label space. A classifier cannot be judged on a label class that is absent from the training reference, absent from the candidate hierarchy or not represented in the external truth mapping. Including such labels only as textual possibilities would create an apparent localization capability without an evaluable denominator. The correct treatment is therefore abstention/out-of-scope reporting. Future work should add cerebellar transcriptomic reference profiles, posterior-fossa label harmonization, MRI-linked posterior-fossa truth and a prospective validation design before any cerebellar or posterior-fossa performance claim is made.

Several layers of transfer separate the training reference from the intended cfRNA use case. First, the reference is tissue RNA-seq, while cfRNA reflects extracellular RNA abundance after cell release, degradation and clearance. Second, cfRNA contains contributions from multiple tissues and blood components, so the brain signal may be diluted. Third, external cfRNA studies often lack patient-level anatomical truth, making apparent localization impossible to validate directly. We therefore separate evidence types: internal Bo2023 LOSO/LOMO measures reference-domain traceability; AHBA measures mapped-label transfer to normal human brain RNA-seq; TCGA/BraTS measures coarse consistency in tumour tissue with MRI-derived labels; and biofluid datasets measure gene coverage and projection feasibility only. The required future validation is a prospective or curated cfRNA cohort with matched imaging or surgical

305 anatomical truth, reported at the same Network/resolution-group/exact-region levels and
306 with abstention for unsupported labels.

307 **Supplementary Tables**

308 **Table S1. Internal validation design**

Validation	Data	Held-out unit	Route tested	Reported endpoints	Denominat or policy
Network LOSO	Bo2023 macaque brain RNA- seq	Single sample	Projected VSD vs logCPM/nat ive VSD Network scoring	Network Top1, Network Top3, median true- rank	All 819 samples
Network LOMO	Bo2023 macaque brain RNA- seq	One monkey	Projected VSD vs logCPM/nat ive VSD Network scoring	Network Top1, Network Top3, median true- rank	All 819 samples
Formal three-tier LOSO	Bo2023 macaque brain RNA- seq	Single sample	Projected- VSD Network beam plus	Network, resolution- group and exact-region	Network n=819; group/exact n=814

			logCPM group/exact rerank	Top1/Top3	
Formal three-tier LOMO	Bo2023 macaque brain RNA- seq	One monkey	Fold-local Network beam plus logCPM group/exact rerank	Network, resolution- group and exact-region Top1/Top3	Network n=819; group/exact n=812

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310 **Table S1a. Ten Network labels and anatomy mapping**

Network	Regions	Resolution groups	Training samples	Classical neuroanatomy mapping
Cingulate gyrus	10	8	39	Medial cingulate cortex; limbic/medial association cortex.
Frontal (agranular frontal motor	7	3	65	Agranular frontal and premotor

areas)				cortex.
Hippocampal formation	1	1	8	Hippocampal/parahippocampal formation.
Lateral Prefrontal Cortex	14	4	108	Dorsolateral and ventrolateral prefrontal association cortex.
Occipital/ Temporal	9	5	69	Visual occipital cortex and occipito-temporal visual association cortex.
Operculum/ Insula	11	2	81	Opercular and insular cortex.
Orbitomedial Prefrontal Cortex (OMPFC)	13	4	103	Orbitofrontal and medial prefrontal cortex.
Parietal, and	11	4	99	Posterior

Parieto-occipital region				parietal and parieto-occipital association cortex.
Subcortical	9	7	54	Basal ganglia, amygdala, thalamic and related subcortical labels in this reference.
Temporal	27	12	193	Temporal association and auditory-related cortex.

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312 **Table S2. Internal validation results**

Dataset	Route	Endpoint	Evaluation	Top1	Top3	Median true-rank	Interpretation
Bo2023 LOSO	Projected VSD	Network	819	58.00%	91.58%	1.0	Supports projected

	Network						d-VSD Network beam
Bo2023 LOMO	Projecte d VSD Network	Network	819	53.72%	91.33%	1.0	Supports donor- level Network beam
Bo2023 LOSO	Formal hybrid	Network	819	58.24%	92.19%	1.0	Primary endpoint across all Network - evaluabl e samples
Bo2023 LOSO	Formal hybrid	Resoluti on group	814	44.47%	72.36%	2.0	Main region- level endpoint among referenc

							e- supporte d samples
Bo2023 LOSO	Formal hybrid	Exact region	814	22.48%	45.33%	4.0	Explorat ory ranking among referenc e- supporte d samples
Bo2023 LOMO	Formal hybrid	Network	819	57.75%	91.21%	1.0	Cross- monkey support
Bo2023 LOMO	Formal hybrid	Resoluti on group	812	41.38%	69.09%	2.0	Main region- level endpoint among referenc e-

							supported samples
Bo2023 LOMO	Formal hybrid	Exact region	812	22.17%	42.36%	5.0	Exploratory ranking among reference- supported samples

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314 **Table S3. External validation design**

Dataset	Sample type	n	Truth label type	Allowed conclusion
AHBA	Human normal brain RNA-seq	242 total; 233 supported; 91 exact-evaluable	Mapped anatomical labels	Cross-species mapped-label validation

TCGA/BraTS	Glioma tissue RNA-seq with MRI-derived labels	65 patients	Human imaging labels	Coarse anatomical consistency only
GSE189919	External count matrix	51 samples	No patient- level anatomical truth	Projection feasibility only

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316 **Table S4. External validation results**

Dataset	Route	Endpoint	Top1	Top3	Conclusion
AHBA	Hybrid	Network	74.68%	94.42%	Strong mapped- label Network support
AHBA	Hybrid	Resolution group	36.26%	67.03%	Moderate group-level support
AHBA	Hybrid	Exact region	24.18%	42.86%	Only exact- evaluable mapped

					labels
AHBA	logCPM baseline	Exact region	17.58%	30.77%	Below hybrid exact Top3
AHBA	Projected VSD only	Exact region	10.99%	29.67%	Below hybrid exact Top3
TCGA/ BraTS	Hybrid	Network	15.38%	40.00%	Coarse consistency only
TCGA/ BraTS	Hybrid	Broad anatomy	13.85%	64.62%	Coarse consistency only
GSE189919	Projection feasibility	Projector gene overlap	15622/2166 8	72.10%	No accuracy claim

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318 **Table S4a. AHBA formal hybrid 95% confidence intervals and random baselines**

Endpoint	Top1, % (95% CI)	Top3, % (95% CI)	Uniform Top3 baseline	Weighted Top3 baseline
Network, n=233	74.68 (68.73- 79.83)	94.42 (90.69- 96.71)	37.5%	56.9%
Resolution	36.26 (27.13-	67.03 (56.86-	20.0%	30.2%

group, n=91	46.51)	75.83)		
Exact region, n=91	24.18 (16.54-33.90)	42.86 (33.18-53.11)	30.0%	32.1%

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320 **Table S4b. Internal weighted random baselines**

Endpoint	Uniform Top3 baseline	Weighted Top3 baseline	Observed Top3
Formal Network LOSO	30.0%	38.4%	92.19%
Formal Network LOMO	30.0%	38.4%	91.21%
Formal resolution-group LOSO	22.0%	6.3%	72.36%
Formal resolution-group LOMO	20.4%	4.4%	69.09%
Formal exact-region LOSO	7.9%	3.2%	45.33%
Formal exact-region LOMO	7.9%	3.2%	42.36%

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322 **Table S5. Figure and table index**

Item	Content	Use in manuscript
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Figure 1	Two-panel workflow and validation summary; Panel B numeric source in Figure1_validation_summary.csv	Main Application Note figure
Table S1	Internal validation design	Documents LOSO/LOMO operations
Table S1a	Ten Network labels and anatomy mapping	Lists all 10 Network classes, region counts, resolution-group counts and classical neuroanatomy mapping
Table S2	Internal validation results	Reports Network, group and exact-region metrics
Table S3	External validation design	Documents label support and allowed conclusions
Table S4	External validation results	Reports AHBA, TCGA/BraTS and GSE189919 outcomes
Table S4a	AHBA formal hybrid 95% confidence intervals and random baselines	Documents uncertainty and baseline context for the main AHBA external

		validation
Table S4b	Internal weighted random baselines	Documents endpoint-specific uniform and weighted random baselines for formal internal validation
Table S5	Figure/table index	Indexes manuscript figure and supplementary tables
Table S6	Claim boundaries	Defines unsupported interpretations and permitted claims
Table S7	Marker selection methodology	Documents marker selection stages, feature counts and fold policy
Table S8	Network-level class F1	Reports class-level precision, recall and F1 for Network endpoints
Table S9	Largest Network Top1 confusion pairs	Summarizes major Network-level error modes and points to repository supplement matrices
Table S10	Named same-field comparator families	Positions cfRNA-BrainTrace relative to

		deconvolution, enrichment and atlas-query comparators
Table S11	Local exact-region fusion- weight sensitivity	Reports available archived lambda sensitivity values for exact-region local fusion

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324 **Table S6. Claim boundaries**

Avoid	Use
Projected VSD creates a new Bo2023 atlas.	Only query profiles are projected for Network beam generation.
Projected VSD is best for exact-region inference.	Projected VSD supports Network beam; logCPM supports downstream reranking.
TCGA/BraTS validates Bo2023 exact regions.	TCGA/BraTS supports coarse anatomical consistency.
GSE189919 validates accuracy.	GSE189919 verifies projection feasibility only.
Exact Top1 is deterministic localization.	Exact outputs are exploratory local candidate rankings.

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Component	Stage	Selection rule	Features	Fold policy
Network global marker panel	Network Top3 beam generation	Fisher-like between- Network / within- Network variance-ratio ranking on Bo2023 training data	200	Rebuilt fold- locally in validation; locked full- reference panel in production
Network pairwise rescue markers	Top1 reordering within retained Network Top3 beam	High-confusion Network-pair discriminative genes	100 per pair	Rebuilt fold- locally in LOMO formal validation; constrained to original beam
Resolution- group local genes	Group reranking inside Network Top3 beam	Local discriminative ranking among candidate groups	200	Fold-local/ sample-local; held-out truth excluded
Exact-region	Exploratory	Nested local	50+100 fused	Fold-local/

local genes	exact-region reranking	Top50/Top100 correlation z- score fusion		sample-local; exact endpoint exploratory
Development marker annotation route	Diagnostic support only	Stable markers with min_consistenc y=0.75, min_effect=0.5 and min_markers_f or_support=8	Top30 per region	Not adopted as main reranking route

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328 **Table S8. Network-level class F1**

Networ k	LOSO support	LOSO precisio n	LOSO recall	LOSO F1	LOMO precisio n	LOMO recall	LOMO F1
Cingula te gyrus	39	0.51	0.59	0.55	0.56	0.62	0.59
Frontal (agranu lar frontal motor	65	0.59	0.54	0.56	0.65	0.68	0.66

areas)							
Hippocampal formation	8	0.57	1.00	0.73	0.67	1.00	0.80
Lateral Prefrontal Cortex	108	0.57	0.47	0.52	0.67	0.43	0.52
Occipital/ Temporal	69	0.71	0.71	0.71	0.73	0.70	0.71
Operculum/ Insula	81	0.42	0.57	0.48	0.39	0.65	0.49
Orbital medial Prefrontal Cortex (OMPFC)	103	0.66	0.59	0.62	0.61	0.50	0.55

Parietal, and Parieto-occipital region	99	0.43	0.78	0.56	0.43	0.74	0.54
Subcortical	54	1.00	0.78	0.88	1.00	0.78	0.88
Temporal	193	0.71	0.44	0.54	0.65	0.44	0.52

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330 **Table S9. Largest Network Top1 confusion pairs**

Validation	True Network	Predicted Network	Count
LOSO	Temporal	Operculum/Insula	47
LOSO	Temporal	Parietal, and Parieto-occipital region	46
LOSO	Orbitomedial Prefrontal Cortex (OMPFC)	Lateral Prefrontal Cortex	21
LOSO	Lateral Prefrontal Cortex	Frontal (agranular frontal motor areas)	21
LOSO	Operculum/Insula	Temporal	15
LOSO	Lateral Prefrontal	Parietal, and	15

	Cortex	Parieto-occipital region	
LOSO	Occipital/Temporal	Parietal, and Parieto-occipital region	14
LOSO	Temporal	Occipital/Temporal	13
LOMO	Temporal	Parietal, and Parieto-occipital region	49
LOMO	Temporal	Operculum/Insula	49
LOMO	Orbitomedial Prefrontal Cortex (OMPFC)	Operculum/Insula	20
LOMO	Lateral Prefrontal Cortex	Frontal (agranular frontal motor areas)	20
LOMO	Lateral Prefrontal Cortex	Parietal, and Parieto-occipital region	17
LOMO	Lateral Prefrontal Cortex	Orbitomedial Prefrontal Cortex (OMPFC)	15
LOMO	Orbitomedial Prefrontal Cortex	Lateral Prefrontal Cortex	14

	(OMPFC)		
LOMO	Operculum/Insula	Temporal	14

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332 Full machine-readable confusion matrices and class-level F1 tables for Network,
 333 resolution-group and exact-region endpoints are available in the repository supplement.

334 **Table S10. Named same-field comparator families**

Tool or family	Primary goal	Difference from cfRNA-BrainTrace
CIBERSORTx and related cell-type deconvolution tools	Estimate cell-type fractions from bulk expression	Cell-type composition, not brain-region hierarchical source ranking
TissueEnrich and tissue-expression enrichment tools	Test tissue enrichment from gene sets or expression signatures	Broad tissue-level inference, not Network to resolution-group to exact-region candidate ranking
Cell-free transcriptome tissue-of-origin studies	Infer broad tissue or cell-type contribution to cfRNA	Important cfRNA context but usually not within-brain atlas localization
Allen/AHBA atlas-query workflows	Map genes or samples to human brain atlas annotations	Atlas-query context; not packaged as a validated app/CLI hierarchical ranker

cfRNA-BrainTrace	Hierarchical brain-origin candidate ranking and resolution-limit auditing	Explicit Top3 beam, three- tier output, confidence diagnostics and shared Streamlit/CLI scoring core
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336 **Table S11. Local exact-region fusion-weight sensitivity**

Route variant	Evaluated n	Exact- region Top1	Exact- region Top3	Median true-rank	Interpretati on
lambda=0.2 5	814	22.48% (183/814)	45.33% (369/814)	4.0	Locked formal route; highest Top1 among evaluated formal LOSO weights
lambda=0.5 0	814	22.36% (182/814)	46.07% (375/814)	4.0	Slightly higher Top3 but lower Top1 than the locked

					route
lambda=0.7	814	22.24%	45.45%	4.0	Similar
5		(181/814)	(370/814)		median
					rank; lower
					Top1 than
					the locked
					route

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338 **Supplementary references**

339 Wilson,E.B. (1927) Probable inference, the law of succession, and statistical inference. J.
340 Am. Stat. Assoc., 22, 209-212.